Predicting Asset Returns via a Structured Strategy Space

A Two-Layer Mathematical Framework for Adaptive Trading

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Outline

1. **The Question** *slide 3*
   What are we predicting, and why is it a well-defined problem?
2. **The Challenge** *slide 4*
   Why standard ML fails on financial time series
3. **Our Solution: Two-Layer Architecture** *slides 5–7*
   Strategy vector space (Layer 1) + sparse linear model (Layer 2)
4. **Validation Design** *slide 8*
   Rolling forward test — no look-ahead bias
5. **Results** *slides 9–10*
   Stage A vs B vs C comparison; why 130 days works
6. **Live System & Conclusion** *slides 11–12*
   Running in production; back to the question

The Question

Central Question

Can we build a mathematically structured prediction system that **identifies which combination of trading signals best predicts near-future returns** — and adapts as market conditions change?

Why this is well-defined:

- Asset: GLD (SPDR Gold ETF) — liquid, well-studied, daily price data
- Prediction target: 130-day forward return (a specific, measurable quantity)
- "Best" = lowest cross-validated prediction error on held-out data
- "Adapts" = model re-estimated every month on a rolling 1-year window

What makes it a math problem: Two objectives — (1) optimize strategy parameters, (2) learn a sparse linear predictor — are composed into a single rolling estimation pipeline.

The Challenge: ML on Financial Time Series

Problem 1: Too many features

- Standard ML needs $n \gg d$ (data $\gg$ features)
- A grid search over strategy parameters produces **hundreds of candidate signals**
- Fitting all of them on one year of data $\Rightarrow$ severe overfitting

Problem 2: Can't just use more data

- Financial data is **non-stationary**: regimes shift
- A longer training window captures outdated market behavior
- *More data* $\neq$ *better model* when the process changes over time

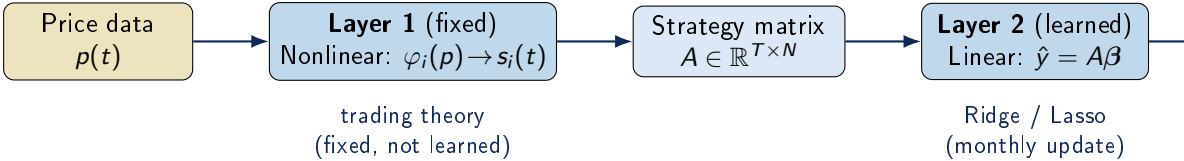
Extend window?
captures old regimes

vs

More features?
overfits on short window

Our solution: use *domain knowledge* to structure the features so the linear layer remains well-conditioned on a 1-year window.

Our Solution: A Two-Layer Mathematical Architecture



Layer 1 — Nonlinear (fixed)

- Converts raw prices into *structured signals*
- Each φ_i is a domain-grounded function (moving averages, volatility bands, ...)
- Equivalent to a fixed first layer of a neural net

Layer 2 — Linear (adaptive)

- $\hat{y} = A\beta$, learned on a rolling 1-year window
- Lasso drives most $\beta_i \to 0$ (automatic feature selection)
- Re-estimated monthly $\Rightarrow$ adapts to regime shifts

The Core Idea: A Strategy Vector Space

Mathematical Foundation of Layer 1

Every trading strategy with parameters θ produces a signal time series $s_\theta(t) \in \{-1, 0, +1\}$ over T days — a **vector in** $\mathbb{R}^T$.

The collection of all such vectors forms a **strategy vector space** $\mathcal{S}$.

How we systematically span $\mathcal{S}$:

1. **Grid search** over θ = sample $\mathcal{S}$ densely
2. **Rank** by performance in selection window
3. **Top-N per family** = basis for the most performant subspace of $\mathcal{S}$
4. **Matrix** A = coordinates of these basis vectors

This is not arbitrary feature engineering.
Each column of A is a *mathematically principled basis vector* derived from the structure of price

4 strategy families span $\mathcal{S}$:

- Adaptive Band
- MA Crossover
- Adaptive Vol. Band
- Fear-Greed C&V

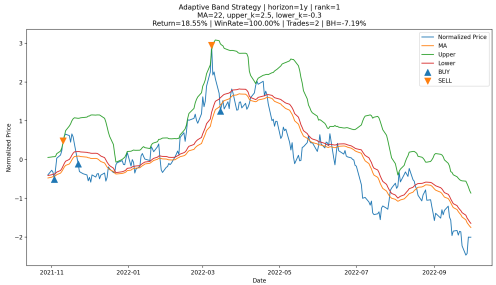
$N = 10$ per family $\Rightarrow$
$A \in \mathbb{R}^{T \times 40}$

Monthly re-selection = subspace adapts to

Two Example Strategies from the Space

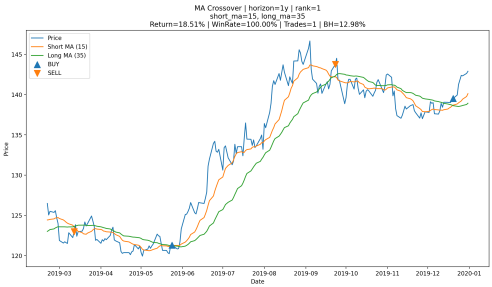
Adaptive Band Strategy

Signal: $+1$ when price crosses above upper band, -1 below lower band



MA Crossover Strategy

Signal: $+1$ when short MA crosses above long MA, -1 when crosses below



Each plot shows: price (top), signal value $s_\theta(t) \in [-1, +1]$ (bottom). Different parameter choices θ produce different vectors in $\mathcal{S}$. Two more families (Adaptive Vol. Band, Fear-Greed) complete the basis.

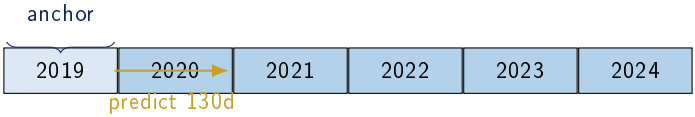
Building Complexity in Three Stages

	Stage A	**Stage B**	**Stage C**
Question	Which single strategy works best now?	What weighted mix maximizes past return?	Which mix best *predicts future* return?
Method	Grid search,		
pick rank-1	Constrained opt.		
over Ax	Ridge / Lasso /		
ElasticNet on A			
Limit	Partial signal	Fixed weights,	
no prediction	—		

All three stages use the **same matrix** A from the strategy vector space.

Stage C asks a fundamentally different question: instead of fitting to past returns, it fits a model to *predict* the next 130-day return.

Validation: Rolling Forward Test (No Look-Ahead)



1. **Anchor date** (e.g., 2020-12-31): grid-search strategy parameters on the past year
2. **Model selection**: fit Ridge/Lasso/ElasticNet on anchor window; pick by CV-MSE
3. **Hold**: deploy signal for 130 days, then recycle capital into the next tranche
4. **Update**: at next month's anchor, re-run steps 1–3

79 anchor dates (2019-12 to 2026-03) covering the full evaluation period.
Forward test: 2020–2024 — all decisions made with data available at the time.

The tranche structure ensures capital is continuously deployed: each day one of the 130 open tranches matures and is reinvested.

Results: Stages A, B, and C Compared

Hold 130d · 1M update · GLD 2020–2024 (Ret / Avg.Exp)

Year	B&H	**A**: Single	**B**: Ensemble	**C**: ML-Guided
2020	+23.9%	+4.4% / 27% exp	+6.4% / 32% exp	+**10.9%** / 57% exp
2021	−6.2%	+0.3% / 25%	+0.3% / 25%	+**0.8%** / 29%
2022	+0.8%	−0.9% / 46%	−2.3% / 42%	−**0.2%** / 31%
2023	+11.8%	+1.6% / 25%	+2.0% / 23%	+**4.0%** / 32%
2024	+27.0%	+5.9% / 21%	+8.0% / 29%	+**16.4%** / 65%
Sharpe (eff.)	—	≈0.5	≈0.7	**1.08**

- Stage C **best in every year** on both return and capital efficiency
- 2024: A(+5.9%) → B(+8.0%) → C(+16.4%) — clear step-up at each stage
- ML-Guided model learns *when* and *how much* to invest, not just which strategy
- 2021 bear year: Stage C still positive (+0.8%) while B&H lost −6.2%

Key Insight: Why Does 130 Days Work Best?

Horizon scan — we tested every prediction target from 5 to 260 days. **130 days (6 months)** was most consistently selected.

The alignment insight

Stage A/B optimize strategy parameters using a **6-month evaluation window** (≈130 trading days). When Layer 2 predicts *the same 130-day horizon*, the features in matrix A are calibrated to exactly the scale the model is trying to predict. **Feature–target alignment**.

Evidence — dedicated model comparison (2020–2024):

Model	2020	2021	2022	2023	2024	Sharpe
h130 (train 130d, hold 130d)	+19%	**+3%**	−1%	+13%	+25%	**1.08**
h45 (train 45d, hold 45d)	+19%	**−18%**	−5%	+6%	+29%	0.33

Values are capital efficiency = return ÷ average exposure.

- h45 catastrophically fails in a bear year (2021: −18% efficiency)
- Shorter horizon targets are noisier → poor generalization in adversarial markets

Stage C in Production: Live Signal

Deployed on Raspberry Pi — runs every trading day at 13:00 PDT

GLD Signal — as of 2026-04-24	
Signal	**BUY (strong)**
Target Weight	1.00 (full position)
Predicted Return	+55.6% over 130 days
Active Model	Ridge
Active Anchor	2026-03-31
Dominant Family	MA Crossover

Pipeline (automated):

1. Fetch latest GLD price (Alpaca API)
2. Re-run Layer 1 with current anchor
3. Layer 2 predicts $\hat{y} \to$ weight w_t
4. Execute tranche order (paper trading)
5. Push signal to GitHub

Infrastructure:

- Raspberry Pi 4 (always-on, low power)
- Alpaca paper trading account
- Systemd timer: runs Mon–Fri only
- Email alert with full report sent daily
- Signal JSON committed to git automatically

Conclusion: Back to the Question

We asked:

Can a structured mathematical framework identify which trading signal combinations best predict near-future returns — and adapt as markets change?

We found: Yes.

- **Two-layer structure** solves the ML challenge: domain-knowledge features (Layer 1) keep the linear layer well-conditioned on a 1-year window
- **Monthly rolling re-estimation** captures regime shifts without sacrificing statistical stability
- **130-day horizon** is the sweet spot: feature–target alignment between Layer 1 optimization and Layer 2 prediction
- **Stage C dominates A and B** in every year of the 5-year forward test (Sharpe 1.08 vs. $\approx$0.5–0.7)
- The system is **live and running** on real market data today

Thank you

Questions welcome

Appendix slides follow for detailed results

Appendix A: Detailed Results — Stage C h130 (top-10)

Return / Avg. Exposure / Capital Efficiency = Return ÷ Exposure

Year	B&H	Strategy Ret	Avg. Exp	Efficiency	vs. B&H
2020	+23.9%	+10.9%	57%	+19%	−13.0%p
2021	−6.2%	+0.8%	29%	+3%	+7.0%p
2022	+0.8%	−0.2%	31%	−1%	−1.0%p
2023	+11.8%	+4.0%	32%	+13%	−7.8%p
2024	+27.0%	+16.4%	65%	+25%	−10.6%p
5yr total	+66.3%	+36.4%	avg 43%	+11.7%	
Sharpe	—	—		1.08	

Interpretation: Strategy return is lower than B&H but uses only 43% of capital on average. Idle capital (57%) can be deployed elsewhere — efficiency per deployed dollar exceeds B&H.

Appendix B: h130 vs h45 Full Comparison (top-10)

Capital Efficiency = Return ÷ Avg. Exposure

	2020	2021	2022	2023	2024	Sharpe
h130 eff	+19%	+3%	−1%	+13%	+25%	**1.08**
h45 eff	+19%	−18%	−5%	+6%	+29%	0.33
h130 ret	+10.9%	+0.8%	−0.2%	+4.0%	+16.4%	
h45 ret	+11.1%	−9.9%	−1.6%	+4.3%	+24.1%	
B&H	+23.9%	−6.2%	+0.8%	+11.8%	+27.0%	

h45 outperforms in bull years (2024: +29% vs +25% eff) but catastrophically fails in bear year (2021: −18% vs +3%). The feature–target misalignment amplifies noise in adversarial conditions.

Appendix C: Update Frequency (Stage C, h130)

Return / Avg. Exposure — more frequent updates generally improve performance

Year	B&H	1M update	2M update	3M update	6M update
2020	+23.9%	+10.9% / 57%	+10.2% / 57%	+9.6% / 51%	+9.7% / 50%
2021	−6.2%	+0.8% / 29%	+0.3% / 35%	+0.6% / 28%	−0.7% / 49%
2022	+0.8%	−0.2% / 31%	+0.9% / 35%	−0.8% / 33%	−1.7% / 44%
2023	+11.8%	+4.0% / 32%	+4.1% / 23%	+3.8% / 40%	+3.4% / 15%
2024	+27.0%	+16.4% / 65%	+13.9% / 60%	+11.1% / 53%	+5.5% / 36%

Monthly (1M) update consistently best — model stays close to current regime. 6M update misses regime shifts: negative in 2021 and 2022.

Appendix D: Strategy Parameter Ranges (Grid Search)

Strategy	Parameter	Range
Adaptive Band	MA window	5, 10, 22, 65, 130, 260 days
	Upper k	-3.0 to $+3.0$ (step 0.1)
	Lower k	-3.0 to $+3.0$ (step 0.1)
MA Crossover	Short MA	5 to 95 days (step 5)
	Long MA	short+5 to 140 days
Adaptive Vol. Band	MA window	5, 10, 22, 65, 130, 260 days
	Vol multiplier	0.5 to 3.0 (step 0.1)
Fear-Greed C&V	Candle threshold	0.3σ to 2.0σ
	Volume threshold	$1.0\times$ to $3.0\times$ avg

Grid search runs in parallel via `joblib` ($4.4\times$ speedup, 20-core CPU). Top-N per family selected by total return in the 1-year selection window.